Assignment 5

```
GPA application almost complete
Lucchese_Anthony_J, Wed Oct 30, 2019, 01:02:15
--------------------------------------------------
no message

Files:
  GPAApplication.java

completed GPA application
Lucchese_Anthony_J, Sat Nov 02, 2019, 23:28:15
--------------------------------------------------
no message

Files:
  GPAApplication.java
```

```
/* GPAApplication.java
**
** Java application that, given an input file containing raw data about
** students, the courses that they took, and the grades they earned in
** those courses, produces a report showing each student's GPA.
**
** CMPS 134 Fall 2019 Prog. Assg. #5
** Authors: Anthony Lucchese, Michael Edwards

** Known defects: GPA itself doesnt work, everything else does, will get it done by thursday
*/
import java.util.Scanner;
import java.io.File;
import java.io.FileNotFoundException;

public class GPAApplication {

   private static Scanner keyboard;  // to read input entered by user
   private static Scanner input;     // to read data from input file

   public static void main(String[] args) throws FileNotFoundException {
      keyboard = new Scanner(System.in);
      String fileName = getStringFromUser("Enter name of input file: ");

     // Establish a Scanner to read from the file with the specified name.
      input = new Scanner(new File(fileName));

      printHeading();

      while (input.hasNextLine()) {
         processStudent();
      }

      System.out.println("\nGoodbye!");
   }

   /* Reads the data pertinent to a single student (name, birthdate, etc.)
   ** using the 'input' Scanner and prints the resulting output record (name,
   ** birthdate (in the required forms), gpa, and a remark, if appropriate.
   */
      private static void processStudent() {

   String studentName = input.nextLine();
   studentName = convertedName(studentName);
   String dateOfBirth = input.nextLine();
   dateOfBirth = convertedDate(dateOfBirth);
   int numberOfCourses = input.nextInt();
   double totalQualityPoints = 0;
   int totalCredits = 0;
      for (int i=1; i<=numberOfCourses; i++) {
      input.next();
      double qualityPoints = 0;
      int credits = input.nextInt();
      String grade = input.nextLine();
      grade = grade.trim();
      qualityPoints = courseQualityPointsOf(credits,grade);
      totalQualityPoints = totalQualityPoints + qualityPoints;
      totalCredits = totalCredits + credits;
      }
      double gpa = gpa(totalCredits,totalQualityPoints);
      String annotation = annotationOf(gpa);
      printStudent(studentName,dateOfBirth,gpa,annotation);
   }



   private static double gpa(int totalCredits, double totalQualityPoints) {
      double gpa = totalQualityPoints / totalCredits;
      return gpa;
   }


   /* Given a student's name, birthdate, GPA, and an annotation, prints
   ** an output record, properly formatted.
   */
```


Commented [PMJ1]: How am I to understand this "authorship" as opposed to "collaboration"?

Commented [PMJ2]: Are you referring to the A&O?

The program works generally but aborts on the third test case.

B

Commented [PMJ3]: Please use indentation.

Assignment 5

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   private static void printStudent(String name, String birthDate,
                                    double gpa, String annotation) {
      System.out.printf("%-25s%11s    %4.2f  %s\n",
                        name, birthDate, gpa, annotation);
   }

   /* Given a name in the "first middle last" format, (with middle name
   ** possibly absent), returns the corresponding name in "last, first" format.
   ** Examples:
   **  "Roger Edwin Goofball" maps to "Goofball, Roger"
   **  "Amy Klunkhead" maps to "Klunkhead, Amy"
   */
   private static String convertedName(String unmapped) {
      String firstName = unmapped.substring(0, unmapped.indexOf(" "));
      String lastName = unmapped.substring(unmapped.lastIndexOf(" "), unmapped.length());
      String name = lastName + ", " + firstName;
      return name;
   }

   /* Given a calendar date expressed in yyyymmdd format, returns the
   ** same date expressed in dd-Mon-yyyy format.
   ** Examples:
   **   "20191016" maps to "16-Oct-2019"
   **   "19870309" maps to "09-Mar-1987"
   */
   private static String convertedDate(String given) {
      String year = given.substring(0, 4);
      String month = given.substring(4, 6);
      String day = given.substring(6, 8);
      month = abbreviationOf(month);
      String dateWithDash = day + "-" + month + "-" + year;
      return dateWithDash;
   }

   /* Given a month number, returns the corresponding 3-character abbreviation.
   ** Each month's abbreviation is the first three letters of its name, with
   ** only the first letter capitalized.
   ** Examples:
   **   3 maps to "Mar"
   **   8 maps to "Aug"
   */
   private static String abbreviationOf(String monthNumber) {
      String result = "";
      if(monthNumber.equals("01")){result = "Jan";}
      else if(monthNumber.equals("02")){result = "Feb";}
      else if(monthNumber.equals("03")){result = "Mar";}
      else if(monthNumber.equals("04")){result = "Apr";}
      else if(monthNumber.equals("05")){result = "May";}
      else if(monthNumber.equals("06")){result = "Jun";}
      else if(monthNumber.equals("07")){result = "Jul";}
      else if(monthNumber.equals("08")){result = "Aug";}
      else if(monthNumber.equals("09")){result = "Sep";}
      else if(monthNumber.equals("10")){result = "Oct";}
      else if(monthNumber.equals("11")){result = "Nov";}
      else if(monthNumber.equals("12")){result = "Dec";}
      return result;
   }

   /* Returns an annotation appropriate to the given GPA.  Specifically, it
   ** returns one of the three strings "Dean's List", "Academic Probation",
   ** or "" (the empty string), according to whether the given GPA value is
   ** at least 3.5, less than 2.0, or neither, respectively.
   */
   private static String annotationOf(double GPA) {
      String result = "";
      if(GPA < 3.5){
         if(GPA < 2.0){
            result = "Academic Probation";}
      } else {result = "Dean's List";}
      return result;
   }

   /* Returns the number of quality points awarded according to the given
   ** letter grade in a course worth the given number of credits.
   */
```

Assignment 5

```
   private static double courseQualityPointsOf(int credits, String letterGrade) {
      double courseQualityPoints = credits * qualityPointsOf(letterGrade);
      return courseQualityPoints;
   }

   /* Returns the number of quality points associated to the given letter grade.
   ** Exhaustive mapping:
   **  "A"  maps to 4.0      "C+" maps to 2.33     "F" maps to 0.0
   **  "A-" maps to 3.67     "C"  maps to 2.0
   **  "B+" maps to 3.33     "C-" maps to 1.67
   **  "B"  maps to 3.0      "D+" maps to 1.33
   **  "B-" maps to 2.67     "D"  maps to 1.0
   */
   private static double qualityPointsOf(String grade) {
      double result = 0.0;
      if (grade.equals("A")) {result = 4.0;}
      else if (grade.equals("A-")) {result = 3.67;}
      else if (grade.equals("B+")) {result = 3.33;}
      else if (grade.equals("B"))  {result = 3.0;}
      else if (grade.equals("B-")) {result = 2.67;}
      else if (grade.equals("C+")) {result = 2.33;}
      else if (grade.equals("C"))  {result = 2.0;}
      else if (grade.equals("C-")) {result = 1.67;}
      else if (grade.equals("D+")) {result = 1.33;}
      else if (grade.equals("D"))  {result = 1.0;}
      else if (grade.equals("F"))  {result = 0.0;}
      return result;
   }

   /* Prints the given prompt and reads/returns a line of data read
   ** using the keyboard Scanner.
   */
   private static String getStringFromUser(String prompt) {
      System.out.print(prompt);
      return keyboard.nextLine();
   }

   /* Prints the heading of the report.
   */
   private static void printHeading() {
      System.out.printf("%-25s%13s  %3s   %10s\n",
                        "Name", "Date of Birth", "GPA", "Annotation");
      printChars('-', 64);
      System.out.println();
   }

   /* Prints the specified character the specified number of times.
   */
   private static void printChars(char ch, int numTimes) {
      for (int i=0; i != numTimes; i++) {
         System.out.print(ch);
      }
   }

}
```

Assignment 5

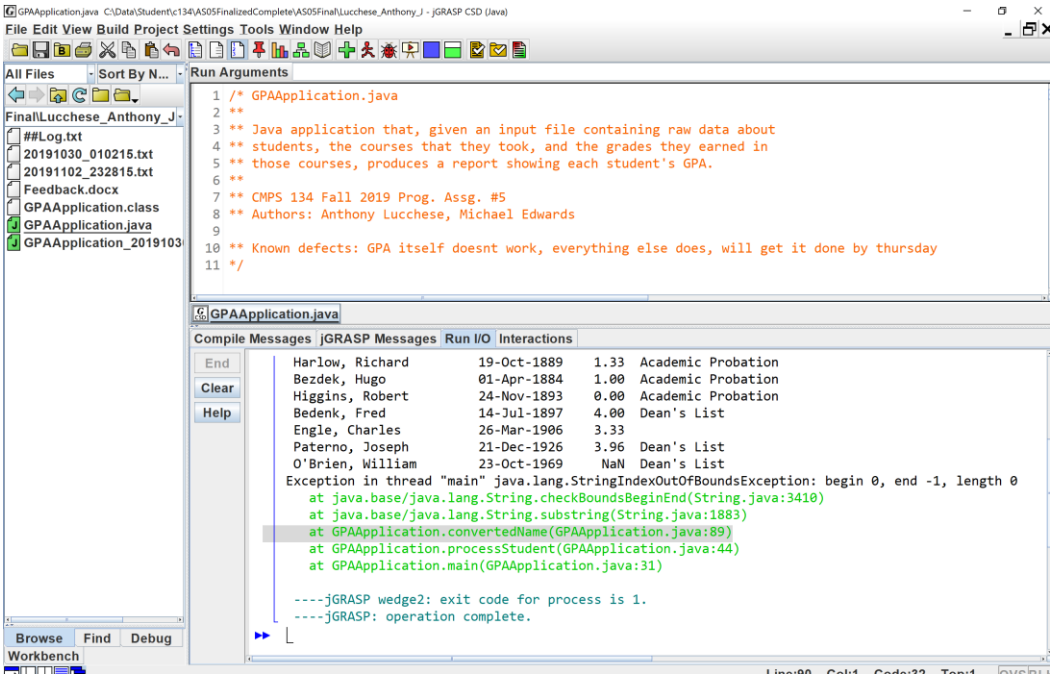
```
Enter name of input file:input.txt
 Name                      Date of Birth  GPA   Annotation
-----------------------------------------------------------------
 Smith, Mary              14-Nov-2000    3.05
 Sharkface, Richard       02-May-2002    3.78  Dean's List
 Glorkhead, Chris         21-Dec-1999    1.79  Academic Probation

Goodbye!
```

```
Enter name of input file:input2.txt
 Name                      Date of Birth  GPA   Annotation
-----------------------------------------------------------------
 Hoskins, George          15-Oct-1864    4.00  Dean's List
 Newton, Sylvanus         04-Dec-1868    3.67  Dean's List
 Boyle, Samuel            30-Oct-1876    3.33
 Golden, William          31-Aug-1868    3.00
 Reed, Daniel             15-Sep-1875    2.67
 Fennell, Thomas          25-May-1875    2.33
 Hollenback, William      22-Feb-1886    2.00
 Hollenback, John         10-Sep-1884    1.67  Academic Probation
 Harlow, Richard          19-Oct-1889    1.33  Academic Probation
 Bezdek, Hugo             01-Apr-1884    1.00  Academic Probation
 Higgins, Robert          24-Nov-1893    0.00  Academic Probation
 Bedenk, Fred             14-Jul-1897    4.00  Dean's List
 Engle, Charles           26-Mar-1906    3.33
 Paterno, Joseph          21-Dec-1926    3.96  Dean's List
 O'Brien, William         23-Oct-1969    3.33
 Franklin, James          02-Feb-1972    3.67  Dean's List

Goodbye!
```

```
Enter name of input file:input3.txt
 Name                      Date of Birth  GPA   Annotation
-----------------------------------------------------------------
 Hoskins, George          15-Oct-1864    4.00  Dean's List
 Newton, Sylvanus         04-Dec-1868    3.67  Dean's List
 Boyle, Samuel            30-Oct-1876    3.33
 Golden, William          31-Aug-1868    3.00
 Reed, Daniel             15-Sep-1875    2.67
 Fennell, Thomas          25-May-1875    2.33
 Hollenback, William      22-Feb-1886    2.00
 Hollenback, John         10-Sep-1884    1.67  Academic Probation
 Harlow, Richard          19-Oct-1889    1.33  Academic Probation
 Bezdek, Hugo             01-Apr-1884    1.00  Academic Probation
 Higgins, Robert          24-Nov-1893    0.00  Academic Probation
 Bedenk, Fred             14-Jul-1897    4.00  Dean's List
 Engle, Charles           26-Mar-1906    3.33
 Paterno, Joseph          21-Dec-1926    3.96  Dean's List
 O'Brien, William         23-Oct-1969     NaN  Dean's List
[ ]
```

Commented [PMJ4]: It aborts at this point.

Assignment 5